

2.3.2 Limits at Infinity

In this section we would like to be able to calculate limits where x is approaching infinity or negative infinity. While the rigorous definitions of what this means can be given, we will stick with intuitive definitions throughout.

Motivation:

What would you expect the following two limits to be: $\lim_{x \rightarrow \infty} \arctan(x)$ and $\lim_{x \rightarrow -\infty} \arctan(x)$?

Definition (Intuitive): Let f be a function defined on some interval (a, ∞) . Then, $\lim_{x \rightarrow \infty} f(x) = L$ means that the values of $f(x)$ can be made arbitrarily close to L by requiring x to be sufficiently large.

Note: One can give an intuitive definition of $\lim_{x \rightarrow -\infty} f(x) = L$ by replacing “sufficiently large” with “sufficiently small”.

Note: We will also consider the cases where the function goes to positive/negative infinity as x goes to positive/negative infinity later in this lecture.

Example 1: Find $\lim_{x \rightarrow \infty} \frac{1}{x}$ and $\lim_{x \rightarrow -\infty} \frac{1}{x}$.

Definition: The line $y = L$ is called a horizontal asymptote of the curve $y = f(x)$ if either $\lim_{x \rightarrow \infty} f(x) = L$ or $\lim_{x \rightarrow -\infty} f(x) = L$.

Note: It is important to understand that a function's graph may cross its horizontal asymptote. All that is important when identifying horizontal asymptotes is the limit.

Note: By the above definition we have that $y = \frac{\pi}{2}$ and $y = -\frac{\pi}{2}$ are horizontal asymptotes of the graph of the inverse tangent function.

Example 2: Find each limit. For limits where x approaches infinity or negative infinity identify any horizontal asymptote(s) of the graph of the corresponding function.

a) $\lim_{x \rightarrow -\infty} \frac{1}{x^2}$

b) $\lim_{x \rightarrow \infty} \sin(x)$

c) $\lim_{x \rightarrow -\infty} e^x$

d) $\lim_{x \rightarrow 2^+} \arctan\left(\frac{1}{x-2}\right)$

e) $\lim_{x \rightarrow \infty} x^2 - 20x + 10$

Part (e) of this example leads us to the following observation.

Fact: When we are computing the limit of a polynomial as x goes to infinity or negative infinity, the highest power term dominates.

Example 3: Find $\lim_{x \rightarrow -\infty} (7x^3 - x^2 + 20x)$

Part (a) of example 2 leads us to the following theorem.

Theorem: If $r > 0$ is a rational number, then $\lim_{x \rightarrow \infty} \frac{1}{x^r} = 0$. Moreover, if x^r is defined for all x , $\lim_{x \rightarrow -\infty} \frac{1}{x^r} = 0$.

This theorem is useful for calculating limits of rational functions and functions close to rational functions when x goes to infinity or negative infinity.

Example 4: Find $\lim_{x \rightarrow -\infty} \frac{4x^2 - x}{2x^3 - 5}$

Example 5: Find each of the following limits.

a) $\lim_{x \rightarrow \infty} \frac{3x + 5}{6x - 8}$

b) $\lim_{x \rightarrow -\infty} \frac{5x^3 - 2x^2 + 1}{1 - 3x}$

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Example 6: Find each of the following limits.

a) $\lim_{x \rightarrow \infty} \frac{\sqrt{2x^2 + 1}}{3x - 5}$

b) $\lim_{x \rightarrow -\infty} \frac{\sqrt{2x^2 + 1}}{3x - 5}$

Example 7: Find $\lim_{x \rightarrow \infty} \sqrt{x^6 + 5x^3} - x^3$. (Hint: Attempt to multiply by 1 in the form of the conjugate)

Example 8:

(a) Find $\lim_{x \rightarrow \infty} e^x$.

(b) Suppose that $f(x) = (1-x)(x+1)^3(x-4)^3$. Find the limit of this function as x goes to infinity, and find its limit as x goes to negative infinity.

(c) Use limits to find all the horizontal asymptotes of the graph of $f(x) = \frac{5e^x + 6}{3 + e^x}$.